

1 SpaceAltimetry.jl: A Julia package for working with 2 NASA satellite (laser) altimetry data

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Software

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6 Summary

7 SpaceAltimetry.jl is a Julia ([Bezanson et al., 2017](#)) package for searching, downloading,
8 and reading spaceborne lidar data from NASA's ICESat, ICESat-2, and GEDI missions. It uses
9 mission-specific granule types and product schemas to expose selected HDF5 variables as lazy
10 [Tables.jl]([Quinn et al., 2026](#))-compatible tables that integrate with the Julia ecosystem.

11 Statement of need

12 The lidar satellite altimetry missions by NASA, ICESat (2003-2010) ([Schutz et al., 2005](#)),
13 ICESat-2 (2019-) ([Markus et al., 2017](#)) and GEDI (2019-) ([Dubayah et al., 2020](#)) are of
14 great use to the scientific community as primary sources of high precision surface height and
15 vegetation characteristics, among others ([Magruder et al., 2024](#)). The data is distributed as
16 product-specific HDF5 granules, with different tracks, variables, quality flags, and coordinate
17 conventions. Research using these datasets commonly requires searching by product, time, and
18 area of interest; downloading authenticated files; selecting variables and tracks; and writing
19 subsets into local tiling schemes. The Python and R ecosystems have packages for (parts
20 of) these workflows, but the Julia ecosystem does not. SpaceAltimetry.jl fills that gap,
21 allowing satellite altimetry data to be easily passed directly into Julia's geospatial and numerical
22 workflows.

23 The package implements the following functionality:

- 24 ▪ **Search:** search queries NASA's Common Metadata Repository (CMR) by mission,
25 product, temporal range, and other metadata.
- 26 ▪ **Download and sync:** download handles authenticated Earthdata downloads for individual
27 granules. For larger jobs, bundled [aria2c](#) support enables parallel downloads, while sync
28 maintains a local folder of granules by scanning existing files and downloading only what
29 is missing.
- 30 ▪ **Variable, attribute, and track selection:** Product-specific schemas define default variables,
31 attributes, and tracks for common workflows. Users can add or replace variables with
32 explicit HDF5 paths and transforms, restrict ICESat-2 and GEDI beams at read time, or
33 use a terminal explorer for unfamiliar HDF5 layouts.
- 34 ▪ **Lazy tables:** `table(g)` returns an `H5Table` for single-track products (ICESat) or a
35 `PartitionedH5Table` for multi-track products (ICESat-2, GEDI), both implementing
36 the `Tables.jl` column-access interface. Columns are read lazily from HDF5, while
37 `_FillValue` and `valid_range` metadata become Julia missing values, flag variables
38 become `CategoricalVectors`, and scalar attributes are broadcast as `FillArrays`.
- 39 ▪ **Filtering and transformations:** Spatial and quality filtering, corrections, and geoid
40 transformations can be chained with Julia's pipe syntax and are specialised to each

41 product. Each operation declares the columns it needs, allowing `SpaceAltimetry.jl` to
42 read the required HDF5 datasets once before materialisation.

43 The package is intended for any Julia workflow using NASA satellite laser altimetry data.
44 `SpaceAltimetry.jl` is especially well suited to repeated regional or global processing from
45 local archives, while keeping outputs compatible with Julia's tabular and geospatial interfaces.

46 State of the field

47 Several tools already support parts of the spaceborne lidar workflow. `SlideRule` (Shean et
48 al., 2023) provides a server-side framework for rapid cloud processing of ICESat-2 data, and
49 `earthaccess` (Barrett et al., 2026) provides a Python package for searching the NASA Earthdata
50 archive.

51 For local ICESat-2 analysis, `icepyx` (Scheick et al., 2023) supports querying, obtaining,
52 and manipulating data in Python. The Photon Research and Engineering Analysis Library
53 (`PhoREAL`) (icesat-2UT, 2026) provides tools and a graphical workflow for ICESat-2 ATL03
54 and ATL08, while `IceSat2R` (Mouselimis, 2026) provides ICESat-2 access in R.

55 For GEDI, `gediDB` (Besnard et al., 2025) provides a Python toolbox for processing and serving
56 GEDI products. The R package `rGEDI` (Silva, 2026) supports GEDI visualisation and processing.

57 `SpaceAltimetry.jl` differs by providing a Julia-native interface and by using the same granule,
58 track-selection, table, and operation concepts across ICESat, ICESat-2, and GEDI. This makes
59 mixed-mission workflows possible without separate mission-specific readers in the user's analysis
60 code, something the Python- and R-based tools above cannot offer Julia users.

61 Software design

62 `SpaceAltimetry.jl` is designed around mission abstractions combined with a geospatially
63 aware but generic HDF5-to-table reader. Scientific, mission-specific code remains in product
64 schemas and operations, while the HDF5 table reader is reusable for arbitrary HDF5 files and
65 provides integration with the Julia ecosystem. This design choice was made because early,
66 hard-coded schemas limited extensibility and adoption.

- 67 ■ **Granules** are the central representation of mission files, storing the identifier, URL, local
68 path, metadata, and footprint where available. Granules are parameterised by product,
69 such as `ICESat2_Granule{:ATL08}` or `GEDI_Granule{:GEDI02_A}`, so Julia's multiple
70 dispatch can use them in table construction and filtering.
- 71 ■ **Variables and attributes** describe how HDF5 datasets become table columns. Product
72 schemas provide curated defaults for paths, element types, scalar attributes, and
73 transforms, while users can inspect schemas, append or replace variables, or define
74 schemas for new products with a small number of methods.
- 75 ■ **The H5Table module** is a generic HDF5-to-Tables.jl layer. It resolves dimensions, flattens
76 compatible multidimensional datasets, handles `_FillValue` and `valid_range` metadata,
77 and encodes flag meanings where available.
- 78 ■ **Lazy operations** separate product semantics from HDF5 reading. Filters and transforms
79 declare the variables they need, so missing input columns can be resolved from the
80 product schema before the table is materialised.
- 81 ■ **Julia integration** is provided through interfaces such as `Tables.jl` and `GeoInterface.jl`.
82 Tables can therefore be passed to existing Julia sinks and geospatial tools, with no-data
83 values represented as missing and repeated or labelled values represented with `Fill` and
84 `Categorical` vectors where appropriate.

Research impact statement

The package grew out of code developed for the first global lowland digital terrain model derived from ICESat-2 satellite lidar (Vernimmen et al., 2020). SpaceAltimetry.jl (then called SpaceLiDAR.jl) was first presented at JuliaCon 2021 (Pronk, M., 2021).

The package was further developed to assess ICESat-2 and GEDI as sources for global terrain modelling (Pronk, Eleveld, et al., 2024). It also underpins DeltaDTM, a global coastal digital terrain model (Pronk, Hooijer, et al., 2024) that used the complete ICESat-2 ATL08 and GEDI L2A archive at the time. It is currently being used for glacier modelling and student research.

AI usage disclosure

AI tools were used during the development of parts of the software and documentation. They helped review implementations, identify bugs or inconsistencies, suggest documentation edits, and improve grammar and spelling, but were not used for the package's overall design, architecture, scientific validation, or interpretation of results.

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References

- Barrett, A., Battisto, C., Bourbeau, J., Carroll, I., Daniels, C., Fisher, M., Kaufman, D., Kennedy, J., Lopez, L., Lowndes, J., Scheick, J., Steiker, A., & Varghese, S. (2026). *Earthaccess*. <https://doi.org/10.5281/zenodo.8365009>
- Besnard, S., Dombrowski, F., & Holcomb, A. (2025). gediDB: A toolbox for processing and providing Global Ecosystem Dynamics Investigation (GEDI) L2A-B and L4A-C data. *J. Open Source Softw.*, 10(113), 8593. <https://doi.org/10.21105/joss.08593>
- Bezanson, J., Edelman, A., Karpinski, S., & Shah, V. B. (2017). Julia: A Fresh Approach to Numerical Computing. *SIAM Rev.*, 59(1), 65–98. <https://doi.org/10/f9wkpj>
- Dubayah, R., Blair, J. B., Goetz, S., Fatoyinbo, L., Hansen, M., Healey, S., Hofton, M., Hurtt, G., Kellner, J., Luthcke, S., Armston, J., Tang, H., Duncanson, L., Hancock, S., Jantz, P., Marselis, S., Patterson, P. L., Qi, W., & Silva, C. (2020). The Global Ecosystem Dynamics Investigation: High-resolution laser ranging of the Earth's forests and topography. *Sci. Remote Sens.*, 1, 100002. <https://doi.org/10/ggjjx8>
- icesat-2UT. (2026). *Icesat-2UT/PhoREAL*. <https://github.com/icesat-2UT/PhoREAL>
- Magruder, L. A., Farrell, S. L., Neuenschwander, A., Duncanson, L., Csatho, B., Kacimi, S., & Fricker, H. A. (2024). Monitoring Earth's climate variables with satellite laser altimetry. *Nature Reviews Earth & Environment*, 5(2), 120–136. <https://doi.org/10.1038/s43017-023-00508-8>
- Markus, T., Neumann, T., Martino, A., Abdalati, W., Brunt, K., Csatho, B., Farrell, S., Fricker, H., Gardner, A., Harding, D., Jasinski, M., Kwok, R., Magruder, L., Lubin, D., Luthcke, S., Morison, J., Nelson, R., Neuenschwander, A., Palm, S., ... Zwally, J. (2017). The Ice, Cloud, and land Elevation Satellite-2 (ICESat-2): Science requirements, concept, and implementation. *Remote Sens. Environ.*, 190, 260–273. <https://doi.org/10/gg3f7c>
- Mouselimis, L. (2026). *Mlampros/IceSat2R*. <https://github.com/mlampros/IceSat2R>

- 126 Pronk, M. (2021). *SpaceLiDAR.jl: Processing ICESat-2 & GEDI satellite LiDAR data*. <https://pretalx.com/juliacon2021/talk/review/UHKBK8NGULYKURYC778VC8E937YFJHYP>
- 127
- 128 Pronk, M., Eleveld, M., & Ledoux, H. (2024). Assessing Vertical Accuracy and Spatial
129 Coverage of ICESat-2 and GEDI Spaceborne Lidar for Creating Global Terrain Models.
130 *Remote Sens.*, 16(13), 2259. <https://doi.org/10.3390/rs16132259>
- 131 Pronk, M., Hooijer, A., Eilander, D., Haag, A., de Jong, T., Voudoukas, M., Vernimmen, R.,
132 Ledoux, H., & Eleveld, M. (2024). DeltaDTM: A global coastal digital terrain model. *Sci*
133 *Data*, 11(1), 273. <https://doi.org/10.1038/s41597-024-03091-9>
- 134 Quinn, J., Kamiński, B., Anthoff, D., Milan Bouchet-Valat, Papp, T. K., Arakaki, T., abhro,
135 Schouten, R., Robinson, N., mathieu17g, Samuel, O., Carlsson, K., Hoffmann, J., Revels,
136 J., ExpandingMan, Hanson, E., Vogt, C., Anthony Blaom, P., Arslan, A., ... Johnson, S.
137 G. (2026). *JuliaData/Tables.jl: V1.13.0*. Zenodo. [https://doi.org/10.5281/ZENODO.](https://doi.org/10.5281/ZENODO.3376179)
138 [3376179](https://doi.org/10.5281/ZENODO.3376179)
- 139 Scheick, J., Leong, W. J., Bisson, K., Arendt, A., Bhushan, S., Fair, Z., Hagen, N. R.,
140 Henderson, S., Knuth, F., Li, T., Liu, Z., Piunno, R., Ravinder, N., Landung, Sutterley, T.,
141 Swinski, J., & Anubhav. (2023). Icepyx: Querying, obtaining, analyzing, and manipulating
142 ICESat-2 datasets. *J. Open Source Softw.*, 8(84), 4912. [https://doi.org/10.21105/joss.](https://doi.org/10.21105/joss.04912)
143 [04912](https://doi.org/10.21105/joss.04912)
- 144 Schutz, B. E., Zwally, H. J., Shuman, C. A., Hancock, D., & DiMarzio, J. P. (2005). Overview
145 of the ICESat Mission. *Geophys. Res. Lett.*, 32(21). <https://doi.org/10/bzbspq>
- 146 Shean, D., Swinski, J. p, Smith, B., Sutterley, T., Henderson, S., Ugarte, C., Lidwa, E., &
147 Neumann, T. (2023). SlideRule: Enabling rapid, scalable, open science for the NASA
148 ICESat-2 mission and beyond. *J. Open Source Softw.*, 8(81), 4982. [https://doi.org/10.](https://doi.org/10.21105/joss.04982)
149 [21105/joss.04982](https://doi.org/10.21105/joss.04982)
- 150 Silva, C. A. (2026). *Carlos-alberto-silva/rGEDI*. <https://github.com/carlos-alberto-silva/rGEDI>
- 151 Vernimmen, R., Hooijer, A., & Pronk, M. (2020). New ICESat-2 Satellite LiDAR Data Allow
152 First Global Lowland DTM Suitable for Accurate Coastal Flood Risk Assessment. *Remote*
153 *Sens.*, 12(17), 2827. <https://doi.org/10/gg9dg6>